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8. The tables show the electronic structures of some Group 1 and Group 2 elements.

Group 1 metal	Electronic structure
sodium	2,8,1
potassium	2,8,8,1

Group 2 metal	Electronic structure
magnesium	2,8,2
calcium	2,8,8,2

(a) (i) Use the information to explain the trend in reactivity down Group 1. [2]

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(ii) Use the information to explain the difference in reactivity of Group 1 and Group 2 elements. [2]

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(b) Sodium reacts with water to produce sodium hydroxide and hydrogen. The equation for this reaction is shown.

$$2Na(s) + 2H_2O(l) \longrightarrow 2NaOH(aq) + H_2(g)$$

Calculate the mass of sodium needed to produce 11.2 g of hydrogen gas. [3]

$A_r(Na) = 23 \qquad A_r(H) = 1$

Mass of sodium = ... g

(c) Group 2 metals react in a similar way with water as Group 1 metals.

The word equation for the reaction of calcium and water is shown.

calcium + water ⟶ calcium hydroxide + hydrogen

Write the balanced symbol equation for this reaction. [2]

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END OF PAPER